

# Tree Height Triangulation Shootout

Measure a tree without climbing it

**At a glance:** Measure a tree without climbing it. Plan about 60 minutes for the core block; 4 teams of 4 students.

## LEARNING OBJECTIVES

BY THE END OF THIS ACTIVITY, EACH STUDENT CAN:

- Explain the core science of this challenge: angles, distance, estimation, remote sensing (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

## MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- 4 paper clinometers (plus 2 spares; GLOBE-style template on cardstock, with a straw and a weighted string)
- String and washers, shared (1 roll string, 1 pack washers; the plumb line for each clinometer)
- 4 twelve-inch rulers (plus 2 spares; to measure each student's eye height)
- 1 long open-reel tape measure (100 ft / 30 m) to pre-mark a fixed standoff distance from the reference tree; the 12 in rulers cannot set this distance
- 4 clipboards (plus 1 spare)
- pencils and a calculator or printed tan table per team, shared

## PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Set up one station per team with the materials above, sorted and ready.
- 02 Stage the scoring sheet and any reference values before testing begins.

**3 No PPE: this is an outdoor measurement activity with no chemicals, water, or soil. Set up sun and hydration support, run the weather/heat-index check, confirm the buddy system and boundaries, and pre-mark the tree-height standoff distance with the long tape.**

- 01 Load the activity slides and ready the projected timer.

### Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Measure a tree without climbing it." Take a quick prediction by show of hands.

### Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

### Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

### Phase 4 Build or solve

0:18 - 0:38

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

### Phase 5 Sight and compute

From the marked standoff distance, each team sights the treetop through the straw, reads and averages the plumb-string angle, measures eye height with the 12 in ruler, and computes tree height = eye height + (distance x tan(angle)).

### Phase 6 Re-measure to improve

Each team takes a second set of sightings (level the eye, average more readings, recheck eye height) and reports whether their estimate moved closer to the reference.

### Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Return the clinometers, string, washers, rulers, the tape, and clipboards to the bin; there is no PPE to remove and nothing to wash.

## TROUBLESHOOTING

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**Problem:** A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

**Problem:** A team's height estimate is far from the reference. **Fix:** Check the clinometer reads 0 when level and 90 straight up, confirm the team sighted the very top from the marked distance (not a guessed distance), average several readings, and confirm they added eye height.

**Problem:** Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

**Problem:** A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

## DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

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Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to angles give height.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

## DEBRIEF QUESTIONS (PICK 2 TO 3)

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- What was your biggest source of error?
- Why does eye height matter in the calculation?
- Where is remote height measurement used in the field?

## SCORING RUBRIC (100 POINTS)

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| SCORING CRITERION | POINTS     |
|-------------------|------------|
| Accuracy          | 50         |
| Method clarity    | 20         |
| Speed             | 20         |
| Teamwork          | 10         |
| <b>Total</b>      | <b>100</b> |

## CLEANUP AND SOURCES

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Return the clinometers, string, washers, rulers, the tape, and clipboards to the bin and collect score slips. The only consumable is the printed clinometer template; there are no surfaces to wipe.

**SOURCES** GLOBE Trees Activities